

Sequential Logic Design. (1)

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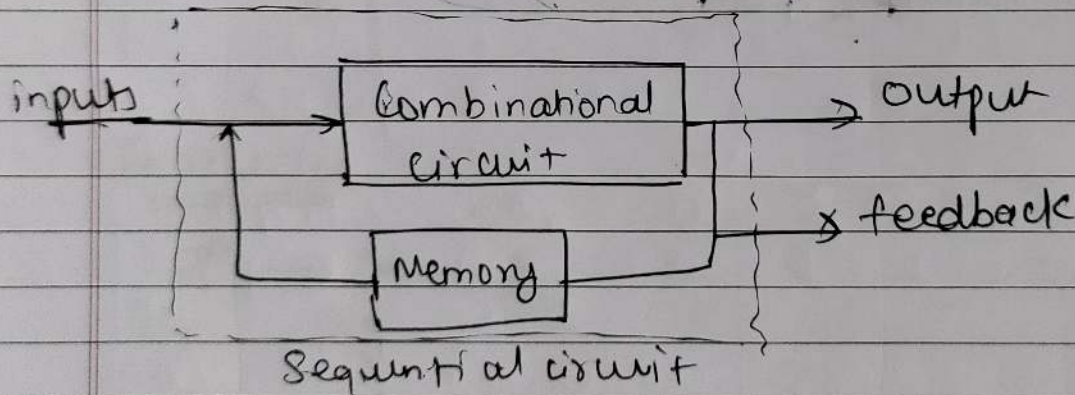
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Intro to Sequential Circuits.

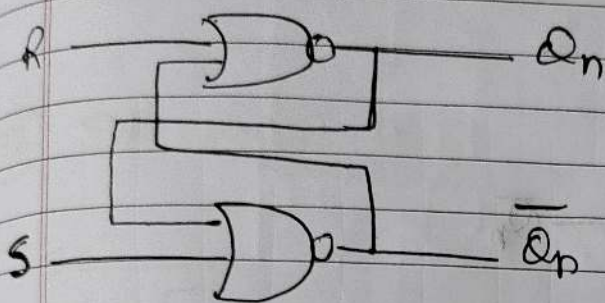
- Sequential circuit is a combinational circuit with memory.
- The output of sequential circuit depends upon present inputs & present state



- The information stored in sequential circuit represents present state
- The present state & present inputs will define output & next state

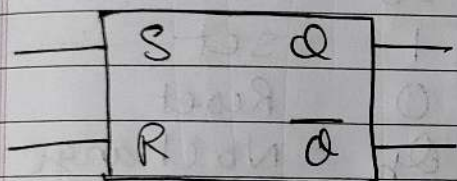
SR Latch

logic circuit



R = Reset
S = set

Logic Symbol



Truth table

	S	R	Q _n	Q _{n+1}
No Change	0	0	0	0
	0	0	1	1
Reset	0	1	0	0
	0	1	1	0
Set	1	0	0	1
	1	0	1	1
S.D.	1	1	0	IP
	1	1	1	IP

(i) $S=0, R=0$

$$Q_{n+1} = 0 + \overline{0 + Q_n} = Q_n$$

(ii) $S=0, R=1$

$$Q_{n+1} = 1 + \overline{0 + Q_n} = 0$$

(iii) $S=1, R=0$

$$Q_{n+1} = 0 + \overline{1 + Q_n} = 1$$

(iv) $S=1, R=1$

$$Q_{n+1} = 1 + \overline{1 + Q_n} = 0$$

$$\overline{Q}_{n+1} = 1 + \overline{1 + Q_n} = 0$$

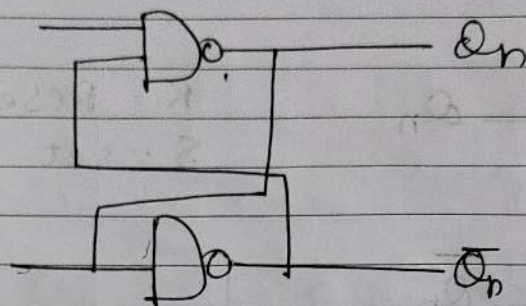
} 3D state

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(171) SR Latch using NAND



S	R	Q_n	Q_{n+1}	
0	0	X	1	
0	1	X	0	set
1	0	X	0	Reset
1	1	X	Q_n	No change

(i) $S = 0$ & $R = 0$

$$\begin{aligned} Q_{n+1} &= 0 \cdot \overline{Q_n} = \overline{0} = 1 \\ \overline{Q_{n+1}} &= \overline{0 \cdot Q_n} = \overline{0} = 1 \end{aligned} \quad \} \neq 0$$

(ii) $S = 1$ & $R = 0$

$$Q_{n+1} = 1 \cdot \overline{Q_n} = \overline{Q_n} = 1$$

(iii) $S = 1$ & $R = 1$

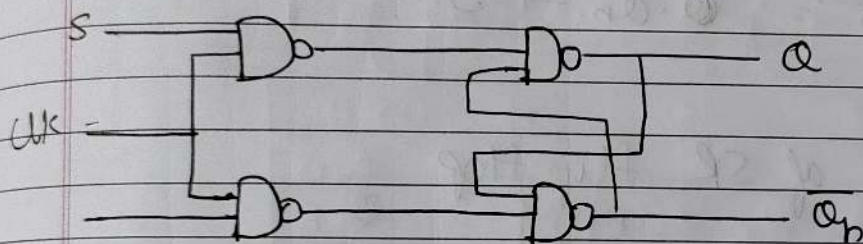
$$\begin{aligned} Q_{n+1} &= 1 \cdot \overline{Q_n} = \overline{Q_n} \rightarrow 0 \\ \overline{Q_{n+1}} &= \overline{1 \cdot Q_n} = \overline{Q_n} = 1 \end{aligned}$$

(iv) $S=1 \quad R=1$

$$Q_{n+1} = \overline{1 \cdot \overline{Q_n}} = Q_n$$

$$\overline{Q_{n+1}} = \overline{1 \cdot Q_n} = \overline{Q_n}$$

(17) SR flip flop



truth table

CLK	S	R	Q_n	Q_{n+1}	
↑	0	0	X	Q_n	No change
↑	0	1	X	0	Reset
↑	1	0	X	1	set
↑	1	1	X	IP.	IP.

(i) $S=0 \quad R=0$

$$Q_{n+1} = \overline{1 \cdot \overline{Q_n}} = Q_n$$

(ii) $S=0 \quad R=1$

$$Q_{n+1} = \overline{1 \cdot \overline{Q_n}} = Q_n \rightarrow 0$$

$$\overline{Q_{n+1}} = \overline{0 \cdot Q_n} = 1 \rightarrow 1$$

(iii) $S=1$ $R=0$
 $Q_{n+1} = 0 \cdot \overline{Q_n} = 0 = 1$

(iv) $S=1$ $R=1$

$Q_{n+1} = 0 \cdot \overline{Q_n}$
 $\overline{Q_{n+1}} = 0 \cdot Q_n = 1$ } Not possible
 IP.

(143) Equation of SR Flip Flop

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	x
1	1	1	x

$Q_n \backslash SR$				
	00	01	11	10
0			x	1
1	1		x	1

$$Q_{n+1} = S + Q_n R$$

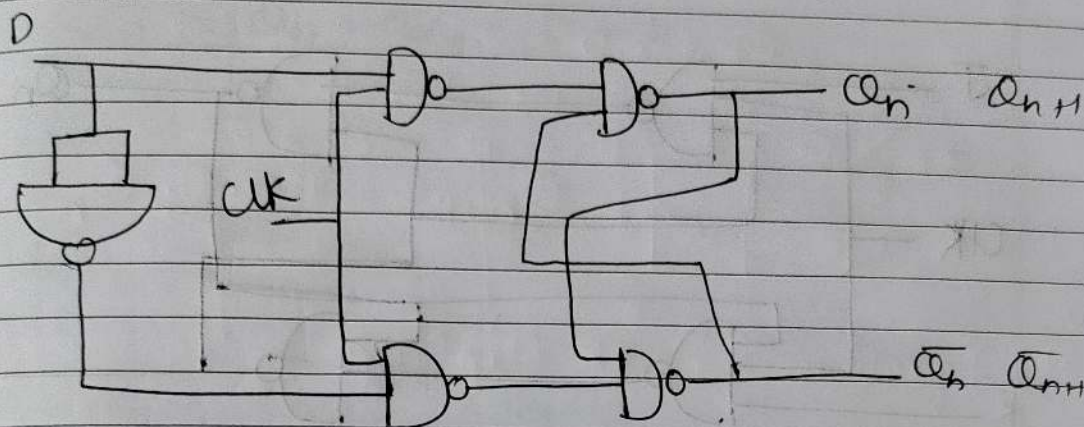
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Data
D Flip Flop



D	Qn	Qn+1
0	X	0
1	X	1

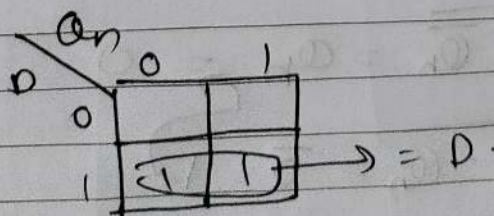
→ Reset
→ Set

(i) $D = 0$

$$Q_{n+1} = 1 \cdot \overline{Q_n} = \overline{Q_n} \rightarrow 0$$

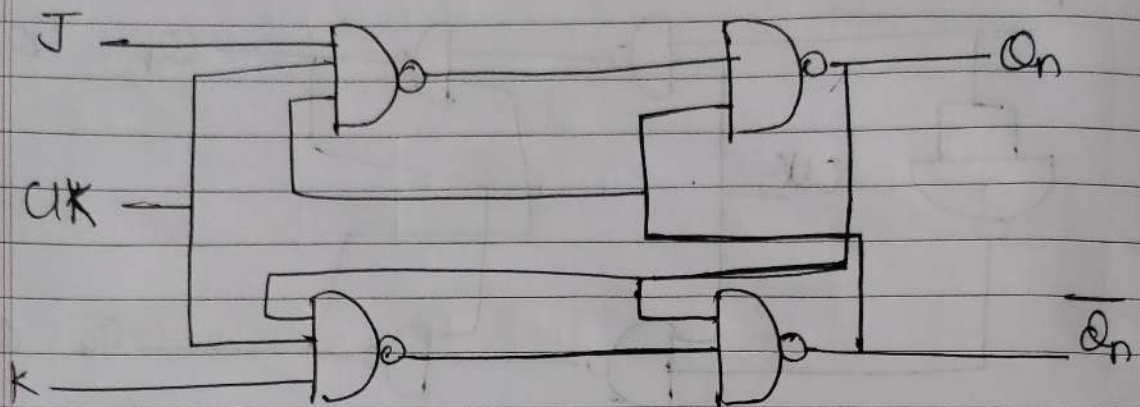
$$\overline{Q_{n+1}} = 0 \cdot Q_n = 0 \rightarrow 1$$

(ii) $\overline{Q_{n+1}} = 0 \cdot Q_n = 1$



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(178) JK flip flop



J	K	Q_n	Q_{n+1}	
0	0	x	Q_n	
0	1	x	0	→ Reset
1	0	x	1	→ Set
1	1	x	$\overline{Q_n}$	→ Toggle

(i) $J=0$ $K=0$

$$Q_{n+1} = 1 \cdot \overline{Q_n} = \overline{Q_n}$$

$$\overline{Q_{n+1}} = 1 \cdot Q_n = Q_n$$

(ii) $J=0$ $K=1$

$$Q_{n+1} = 1 \cdot \overline{Q_n} = \overline{Q_n} \rightarrow 0$$

$$\overline{Q_{n+1}} = \overline{Q_n} \cdot Q_n = 0 \rightarrow 1$$

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(iii) $J=1 \quad K=0$

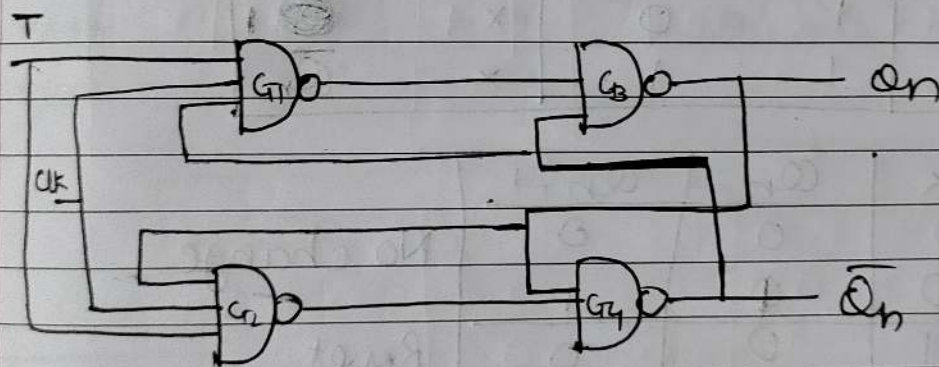
$$Q_{n+1} = Q_n \cdot \overline{Q_n} = \overline{0} = 1$$

(iv) $J=1 \quad K=1$

$$\overline{Q_{n+1}} = \overline{Q_n \cdot Q_n} = \overline{0} = 1$$

$$Q_{n+1} = \overline{Q_n \cdot 1} = \overline{Q_n}$$

(176) T Flip Flop (Toggle)



T	Q_n	Q_{n+1}
0	X	Q_n
1	X	$\overline{Q_n}$

(i) $T=0$

No change $Q_{n+1} = \overline{1 \cdot \overline{Q_n}} = \overline{\overline{Q_n}} = Q_n$
 $Q_{n+1} = \overline{1 \cdot Q_n} = \overline{Q_n}$

(ii) $T=1$

$$G_1 = Q_n \quad G_2 = \overline{Q_n}$$

when $Q_n = 0$ $Q_{n+1} = \overline{0 \cdot \overline{Q_n}} = \overline{0} = 1$
 $Q_n = 1$ $\Rightarrow \overline{Q_n} = 0$

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$$G_y(0/p) = \overline{Q_{n+1}} = \overline{0 \cdot Q_n} = 1$$

$$Q_{n+1} = 0$$

	T	0	1
Q _n	0		1
	1	1	

$$Q_{n+1} = T \oplus Q_n$$

(147) Equation for JK Flipflop.

clk	J	K	Q _n	Q _{n+1}
↓	x	x	x	Q _n
↑	0	0	x	Q _n
↑	0	1	x	0
↑	1	0	x	1
↑	1	1	x	$\overline{Q_n}$

J	K	Q _n	Q _{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

No change

Reset

Set

Toggle

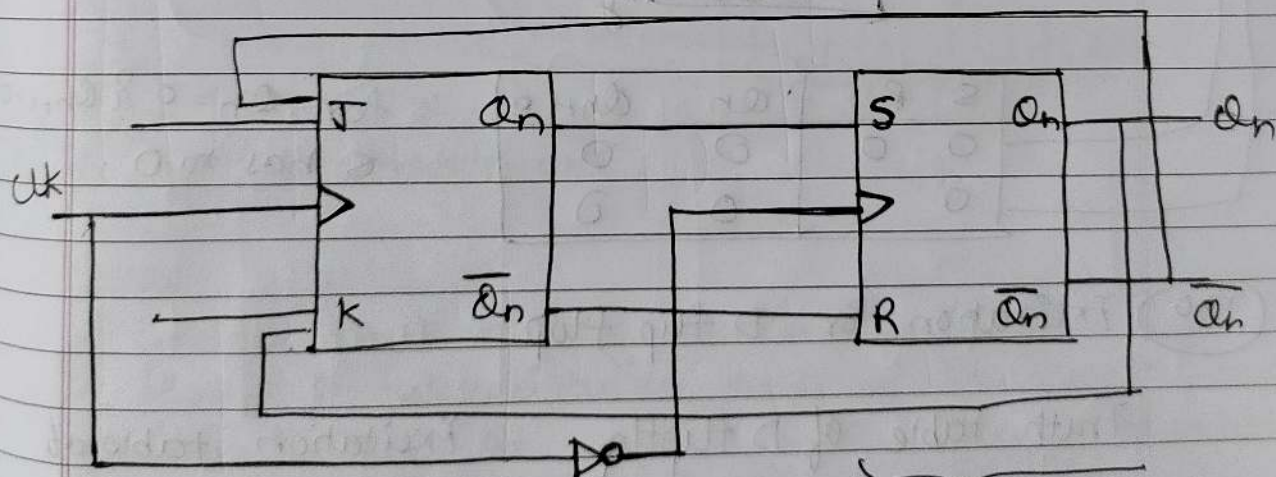
K map Q_{n+1}

$Q_n \backslash JK$	00	01	10	11
0			1	1
1	1			1

$$Q_{n+1} = \overline{Q_n}J + Q_nK$$

178 Race Around condition (Master slave JK FlipFlop)

For JK FlipFlop if $J=K=1$ & if clock is too long then state of FlipFlop keep on toggle which leads to uncertainty in determining output state of FlipFlop. This condition is called race around condition.



JK master

SR slave
(JK can also be used)

(179) Excitation table for SR flip flop

Just for understanding

for a particular combination of present state and next state what may be the mandatory input that we need to provide is given by the excitation table

Truth
~~table~~ table of SR
Flip Flop

Excitation table for
SR Flip Flop

S	R	Q_n	Q_{n+1}
0	0	$x \rightarrow 0$	Q_n
0	1	$x \rightarrow 0$	0
1	0	$x \rightarrow 0$	1
1	1	$x \rightarrow 0$	1

Q_n	Q_{n+1}	S	R
0	0	0	x
0	1	1	0
1	0	0	1
1	1	x	0

for understanding

S	R	Q_n	Q_{n+1}
0	0	0	0
0	1	0	0

$\therefore$ For $Q_n = 0$ & $Q_{n+1} = 0$
S has to 0.

(180) Excitation for D flip flop.

Truth table of D flip flop

D	Q_n	Q_{n+1}
0	$x \rightarrow 0$	0
1	$x \rightarrow 0$	1

Excitation table of
D flip flop.

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

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(81) Truth table of JK

Truth table of JK FlipFlop

J	K	Q_n	Q_{n+1}
0	0	$X \rightarrow 1$	$Q_n \rightarrow 1$
0	1	$X \rightarrow 1$	0
1	0	$X \rightarrow 0$	1
1	1	$X \rightarrow 1$	$Q_n \rightarrow 0$

Excitation table of JK flipflop

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	0	X
1	0	X	1
1	1	X	0

(82) Truth table of T flip flop

T	Q_n	Q_{n+1}
0	$X \rightarrow 1$	$Q_n \rightarrow 1$
1	$X \rightarrow 1$	$Q_n \rightarrow 0$

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

(83) FlipFlop conversions.

Steps

- Consider truth table for destination flipflop & extend it as excitation table of source flipflop
- Draw Kmap for source flipflop input variables, it will provide expression for conversion
- Draw circuit according to the kmap expression

Conversion $\rightarrow$ SR $\rightarrow$ DSR $\rightarrow$ JKSR $\rightarrow$ TJK $\rightarrow$ TJK $\rightarrow$ DT $\rightarrow$ DD $\rightarrow$ T

(184) SR flipflop to D flipflop

Truth table for D.

D	Q_n	Q_{n+1}
0	X	0
1	X	1

Excitation table for SR

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

D	Q_n	Q_{n+1}	S	R
0	0	0	0	X
0	1	0	0	1
1	0	1	1	0
1	1	1	X	0

Truth table of D

Excitation table of SR flipflop

K-map for S.

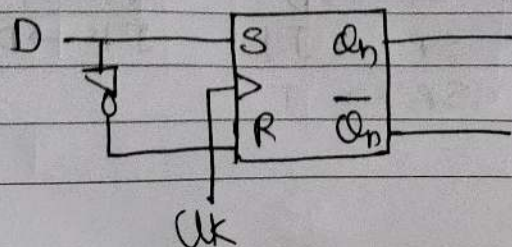
$Q_n \backslash D$	0	1
0	0	0
1	1	X

$$S = D$$

K-map for R

$Q_n \backslash D$	0	1
0	0	0
1	X	0

$$R = 0$$



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(155) SR to JK

Truth table of JK

J	K	Q_n	Q_{n+1}
0	0	x	Q_n
0	1	x	0
1	0	x	1
1	1	x	$\overline{Q_n}$

Excitation table for SR

Q_n	Q_{n+1}	S	R
0	0	0	x
0	1	1	0
1	0	0	1
1	1	x	0

J	K	Q_n	Q_{n+1}	S	R
0	0	0	0	0	x
0	0	1	1	x	0
0	1	0	0	0	x
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	x	0
1	1	0	1	1	0
1	1	1	0	0	1

Truth table
for JK Flip
Flop

Excitation table for
SR

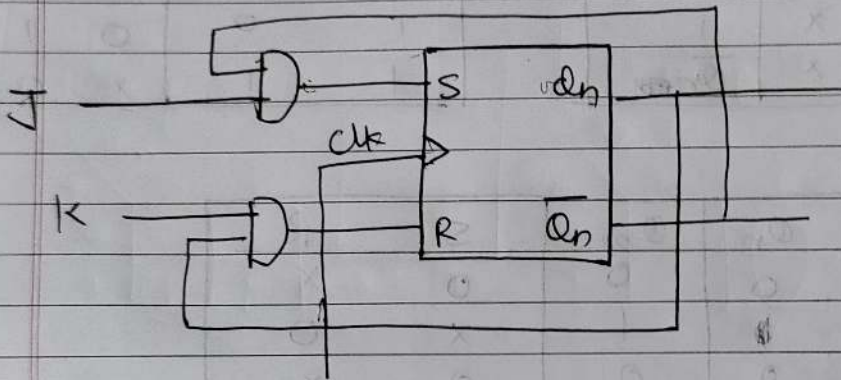
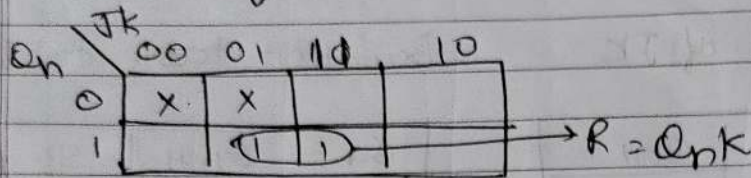
JK Kmap for S

Q_n	00	01	11	10
0			1	1
1	x			x

$$S = \overline{Q_n} \cdot J$$

(15)

K-map for f



(186)

SR to T.

Truth table for T.

T	Q _n	Q _{n+1}
0	x	Q _n
1	x	$\overline{Q_n}$

Excitation table for SR

Q _n	Q _{n+1}	S	R
0	0	0	x
0	1	1	0
1	0	0	1
1	1	x	0

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T	Q _n	Q _{n+1}	S	R
0	0	0	0	x
0	1	1	x	0
1	0	1	1	0
1	1	0	0	1

Truth table for T

Excitation for
SR

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K map for S

Q_n S

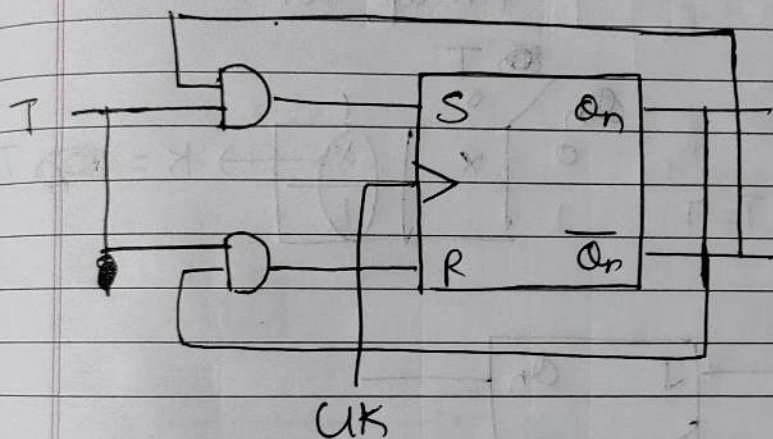
T	0	1
0		X
1	1	

$$S = T \bar{Q}_n$$

K map for R

T	0	1
0	X	
1		1

$$R = T Q_n$$



167 JK to T.

Truth table of T

T	Q_n	Q_{n+1}
0	X	Q_n
1	X	$\bar{Q}_n$

Excitation table of JK

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

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T	Q_n	Q_{n+1}	J	K
0	0	0	0	X
0	1	1	X	0
1	0	1	1	X
1	1	0	X	1

Truth table of D

Excitation of JK

K map for J

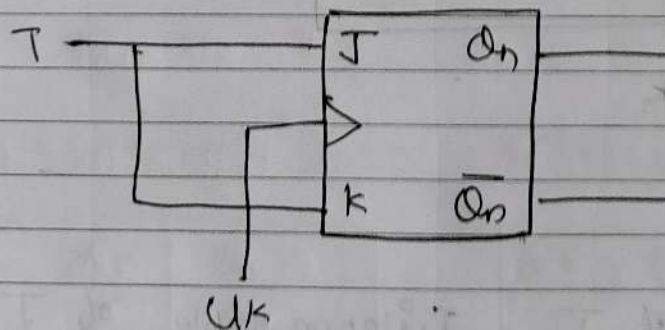
Q_n	0	1
T	0	X
1	1	X

$\rightarrow J = 1$

K map for K

Q_n	0	1
T	X	X
1	X	1

$\rightarrow K = 1$



(188) JK to D

Truth table of D

D	Q_n	Q_{n+1}
0	X	0
1	X	1

Excitation table of JK

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

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D	Q_n	Q_{n+1}	J	K
0	0	0	0	*
0	1	0	x	1
1	0	1	1	x
1	1	1	x	0

Truth table
of D

Excitation of JK

Kmap for J

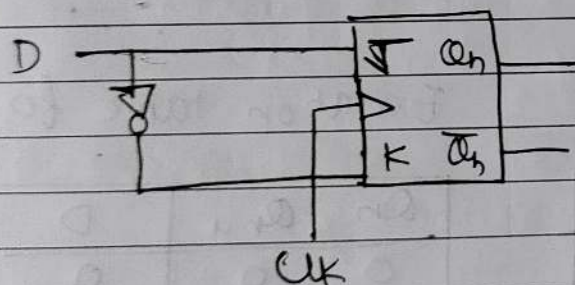
Q_n	0	1
D	0	1
0		1
1	x	x

$\rightarrow J = D$

Kmap for K

Q_n	0	1
D	0	1
0		x
1		1

$\rightarrow K = \bar{D}$



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T to D.

Truth table of D.

D	Q_n	Q_{n+1}
0	x	0
1	x	1

T

Excitation for T.

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

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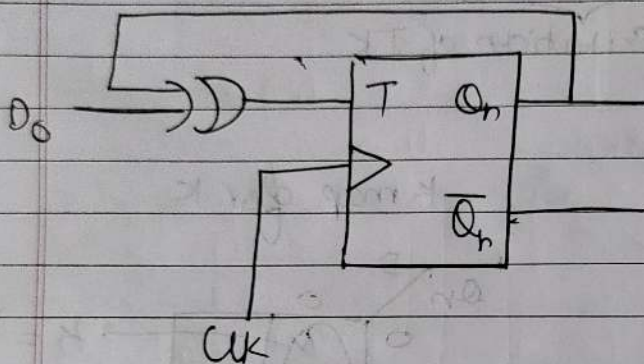
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D	Q_n	Q_{n+1}	T
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

K map of T

Q_n	0	1
D	0	1
1	1	0

$$T = D \oplus Q_n$$



(191)

(190) D ~~flip~~ to T.

Truth table of T

T	Q_n	Q_{n+1}
0	X	Q_n
1	X	$\overline{Q_n}$

Excitation table for D

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

T	Q_n	Q_{n+1}	D
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

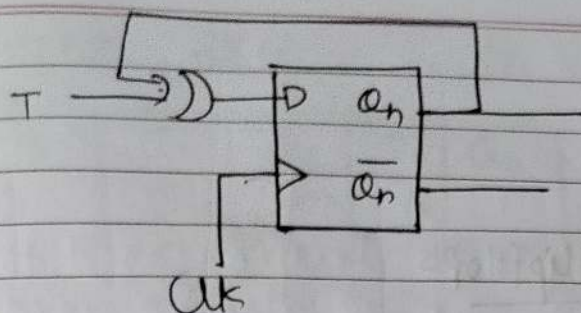
Truth table of D

Excitation table of D.

K map for D

Q_n	0	1
T	0	1
1	1	0

$$D = T \oplus Q_n$$

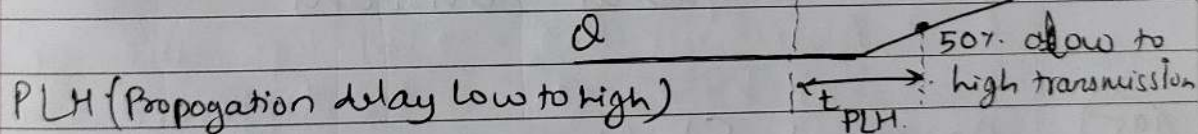


(91) Propagation Delay

It is time required to change the output after application of input. We are having two types of propagation delay

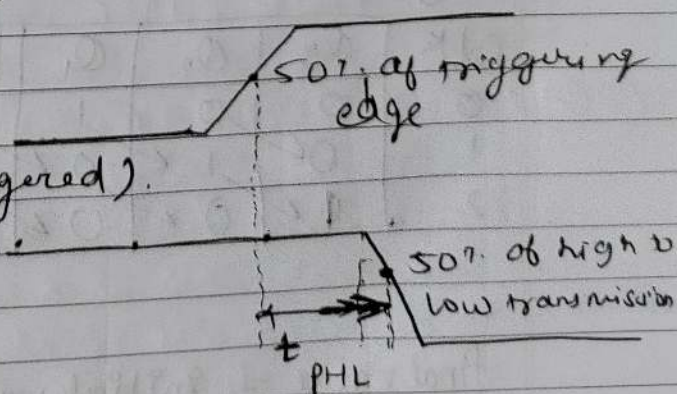
i.e. propagation delay t_{PLH} :- CLK

(Clock is -ve edge triggered)



Propagation delay t_{PHL} :-

(Clock is +ve edge triggered)



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① Shift Registers (Group of Flip Flops)

- parallel to serial conversion
- Serial to parallel conversion
- Data ~~to~~ Storage — Serial
 └─ Parallel

(2) Counters.

- count the no. of pulses
- frequency divisions

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Universal Shift register

Left Shift

Q_k	Q_3	Q_2	Q_1	Q_0
0	0	0	1	0
1	0	1	0	0
2	1	0	0	0

Final value = initial value $\times 2^n$

$n = \text{no. of clocks}$

Right Shift

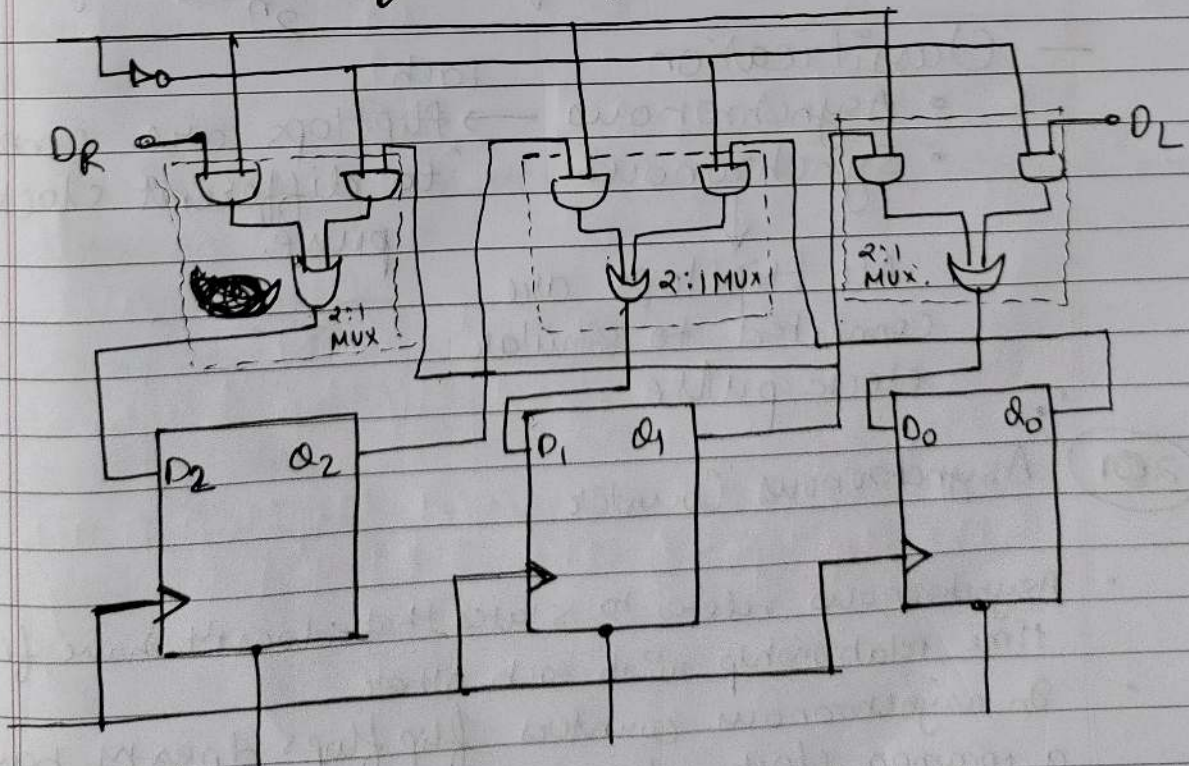
CK	Q_3	Q_2	Q_1	Q_0
0	0	0	1	1
1	0	0	0	1

$$\text{Final value} = \text{Quotient} \left(\frac{\text{Initial value}}{2^n} \right)$$

Not in playlist

Bidirectional Shift Register

Bidirectional Shift register can shift no. either to left side or to right side
 → for understanding



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200 Counters

- Counter is a digital device used to count no. of pulses & it can ~~be~~ also be used as frequency divider
- Counter can count in two ways i.e.
 - (i) up count (0, 1, 2, ..., N)
 - (ii) down count (N, N-1, ..., 1, 0)
- Present count of the counter represents state of counter
- Counter contains set of flip flops. A n-bit counter will have N flip flops & 2^n states
- Each state frequency = $\frac{\text{total frequency}}{2^n}$

Classification

- Asynchronous → Each flip flops are connected to different clock pulse
- Synchronous →

↓
All flip flops are connected to similar clock pulse

201 Asynchronous Counter

- Asynchronous refers to states that doesn't have fixed time relationship with each other
- In asynchronous counters flip flops doesn't have a common clock pulse
- So, their states doesn't change exactly at the same time

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High

U

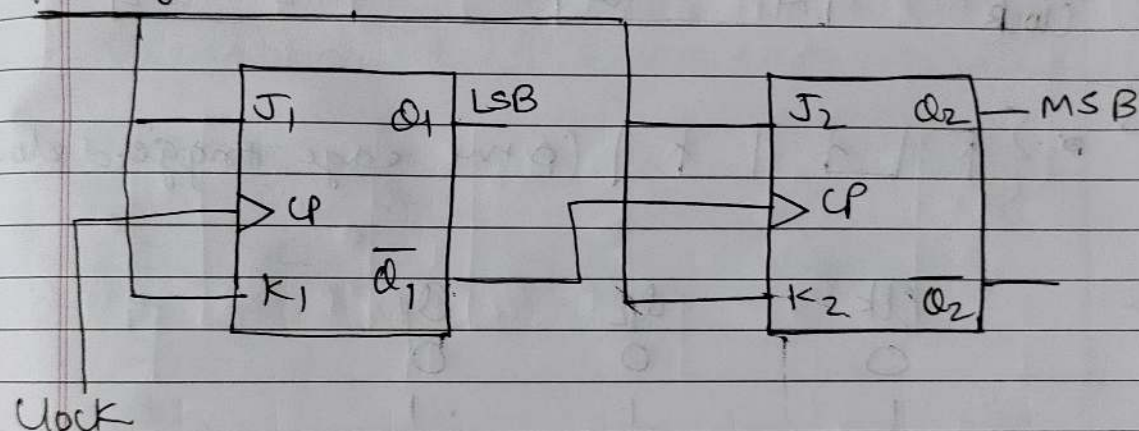
Just know

- 1st Flip Flop has clock, output of each flip flop acts as a clock for the next flip flop in the counter.

Example:- Ripple counter

2 bit + Ripple up counter

High voltage



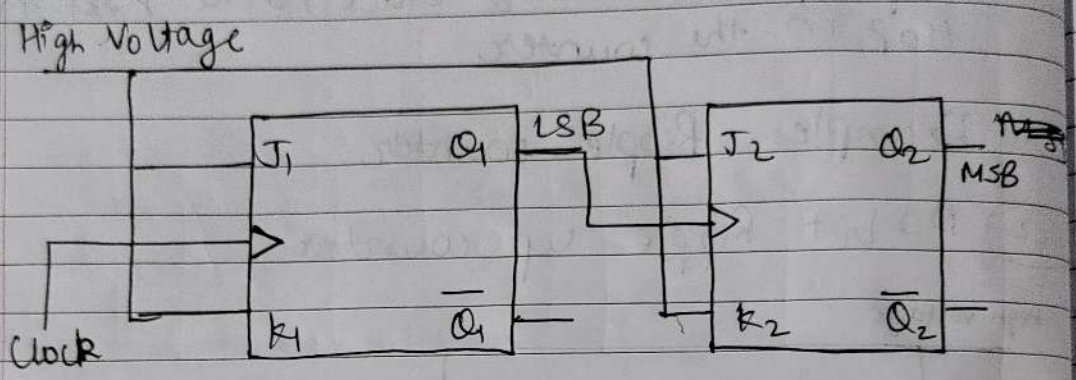
for a 2 bit counter there are 2^n i.e. $2^2 = 4$ states

Just know

Clock - ?

clk	Q_2	Q_1
0	0	0
1	0	1
2	1	0
3	1	1
4	0	0

Q 2 bit Ripple down counter

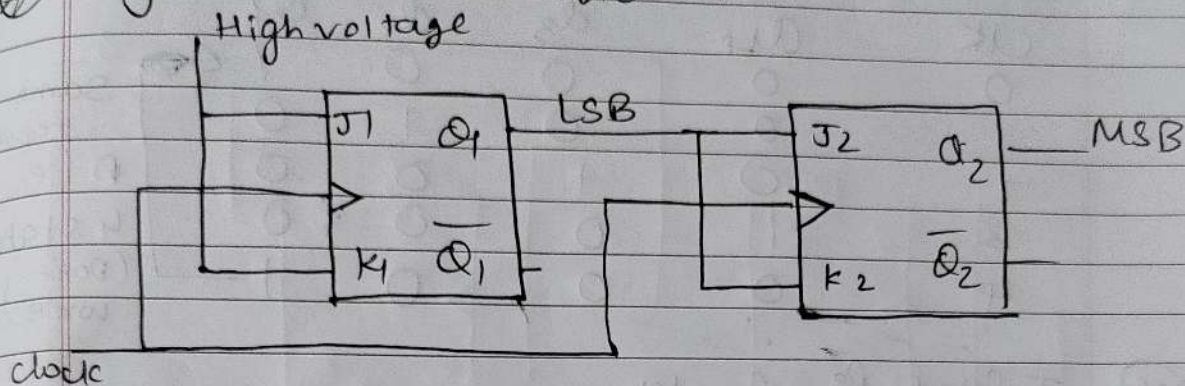


Just know (for the edge triggered clock)

clk	Q2	Q1
0	0	0
1	1	1
2	1	0
3	0	1
4	0	0

{ for the edge triggered flipflop 1 activates the flipflops
for the edge triggered flipflop 0 activates the flipflops }

Synchronous Counter

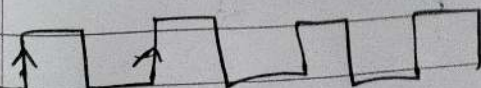
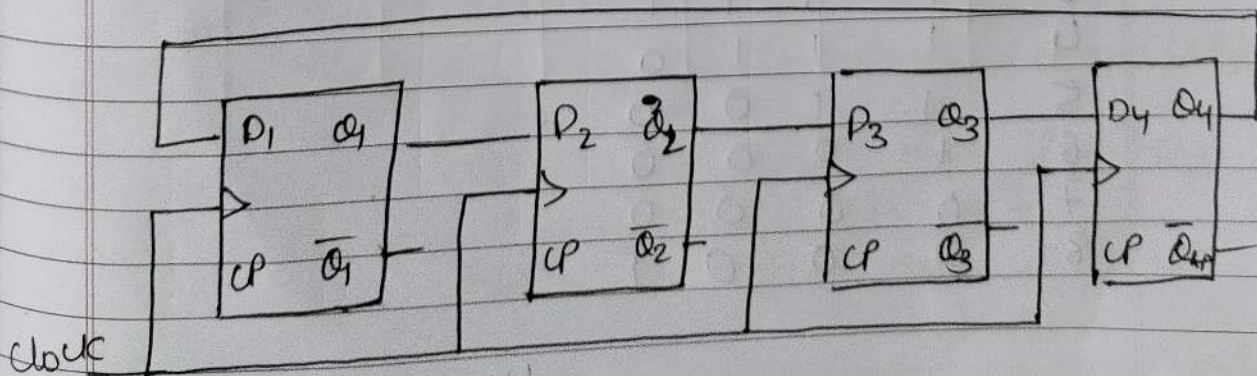


① (the edge triggered clock)

clk	Q_2	Q_1
0	0	0
1	0	1
2	1	0
3	1	1
4	0	0

205 Ring counter @ Shift register counter

Ring counter doesn't have 2^n states
it just have n states.



(27)

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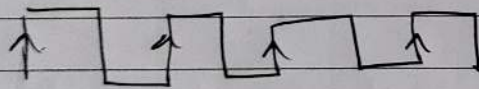
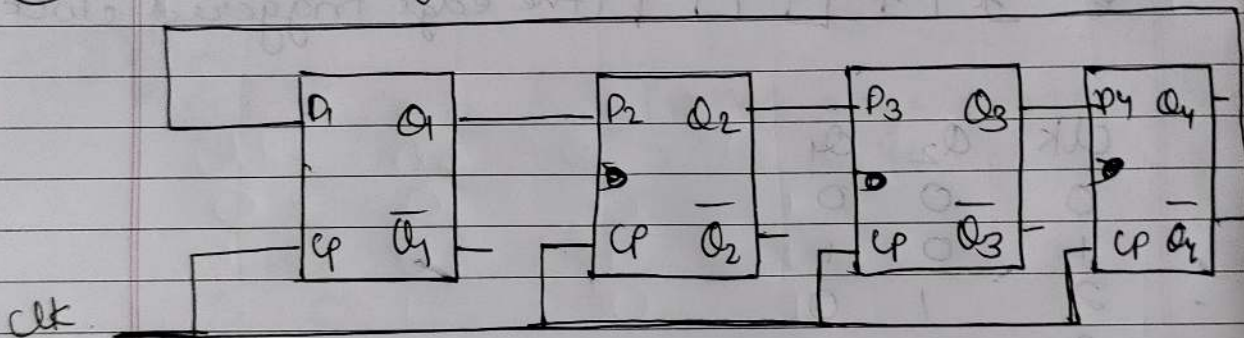
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clk	Q_4	Q_3	Q_2	Q_1
0	0	0	0	1
1	0	0	1	0
2	0	1	0	0
3	1	0	0	0
4	0	0	0	1

Same $\therefore$ require 4 states (Don't write)

(206) Twisted Ring Counter



clk	Q_4	Q_3	Q_2	Q_1
0	0	0	0	1
1	0	0	1	0
2	0	1	1	1
3	1	1	1	1
4	1	1	1	0
5	1	1	0	0
6	1	0	0	0
7	0	0	0	0
8	0	0	0	1

total no. of states are $2N$

(207)

(208)

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- ~~Ring state freq.~~ Ring counter. Each state
$$f_{\text{req}} = \frac{f}{N}$$
- Asynchronous total states = 2^n .
Each state $f_{\text{req}} = \frac{f}{2^n}$

(207)

(208)

(209)

(210)

Modulus of counter

The no. of states passed by the counter before going to its initial state is called as modulus of counter.

(211)

(212)

(213)

Asynchronous $\odot$ direct inputs.

(214)

~~Asynchronous counter~~

- Asynchronous inputs can set or reset the flipflop regardless of the states of clock plus
- Typically they are named as Clear (CLR), preset (PRR)
- When preset input is active then flipflop is ~~set~~ going to set ($Q = 1, \bar{Q} = 0$)
- When clear input is active then flipflop is going to reset ($Q = 0, \bar{Q} = 1$). both are on then we will get invalid state of flipflop.

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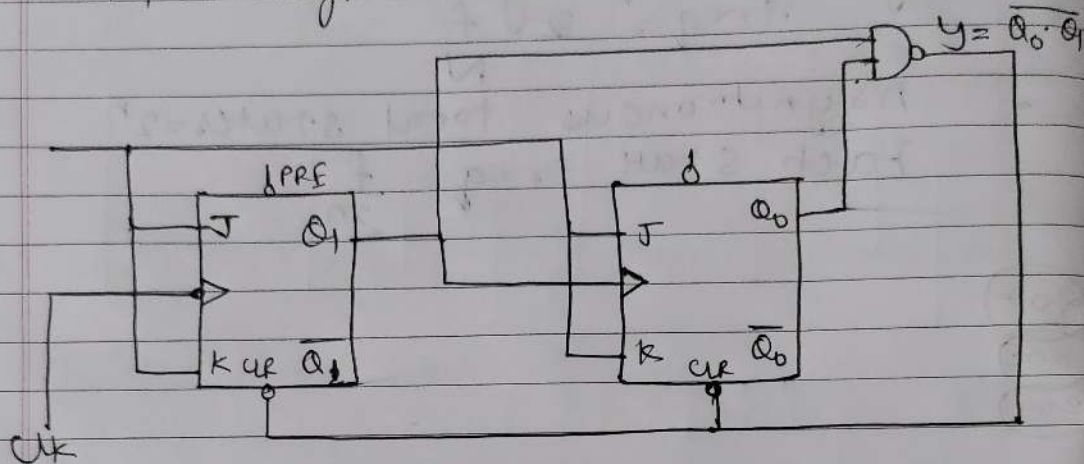
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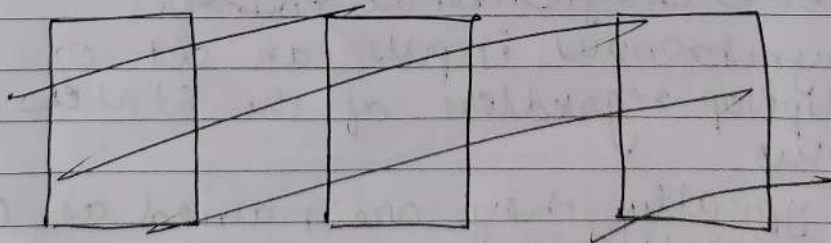
MOD 3 Asynchronous counter



clk	Q_0	Q_1	$Y = \overline{Q_0} \cdot Q_1$
0	0	0	1
1	0	1	1
2	1	0	1
3	1	1	0

(215)

MOD 12 counter using T flipflop



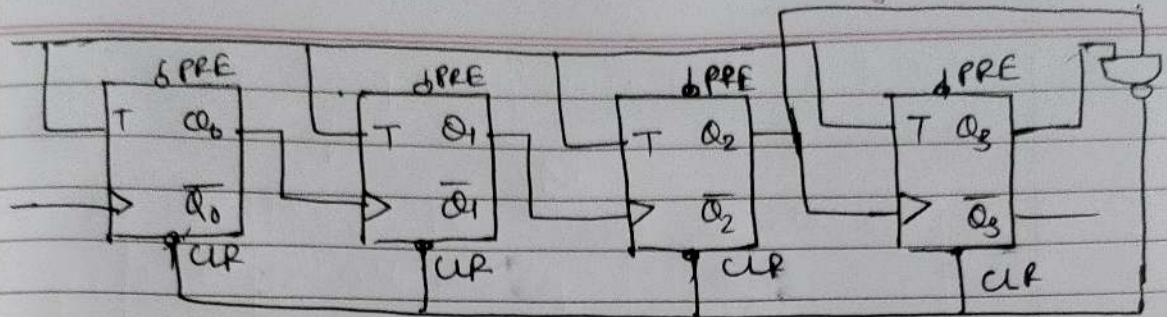
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CLK	Q ₃	Q ₂	Q ₁	Q ₀
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0

Stops the counting at 12 pulses.